

Total No. of Printed Pages:02

SUBJECT CODE NO: - RNP-8045
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (Civil Engg.) (NEP) Sem - IV
Examination May/June-2026
Concrete Technology

[Time: 3:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

Q.1 A) Choose the right option for the multiple choice questions given below. 10

- 1) Initial setting time of Ordinary Portland Cement should not be less than:
 - a) 10 minutes
 - b) 20 minutes
 - c) 30 minutes
 - d) 60 minutes
- 2) Fineness modulus is used to determine:
 - a) Strength of cement
 - b) Gradation of aggregates
 - c) Workability
 - d) Setting time
- 3) Flakiness index relates to:
 - a) Shape of aggregates
 - b) Strength of cement
 - c) Water quality
 - d) Curing process
- 4) Bleeding in concrete refers to:
 - a) Loss of cement
 - b) Upward movement of water
 - c) Cracking
 - d) Segregation of aggregates
- 5) Self-compacting concrete requires:
 - a) Manual compaction
 - b) Vibrators
 - c) No compaction
 - d) High temperature curing

B) Answer any 5 from the following questions. 10

- a) Define standard consistency of cement.
- b) What are cementitious materials?
- c) Name any two supplementary cementitious materials.
- d) Define water absorption.
- e) What is fineness of cement?
- f) Define bulk density.
- g) What is plasticizer?

- Q.2** a) Explain production and properties of cement **05**
 b) Describe standard specifications of cement. **05**

- Q.3** a) Explain properties of water used in concrete. **05**
 b) Discuss properties and tests of coarse aggregates. **05**

- Q.4** a) Explain principles of concrete mix design. **05**
 b) Describe IS method of concrete mix design. **05**

- Q.5** a) Explain cohesiveness and bleeding in fresh concrete. **05**
b) Describe methods of measuring workability. **05**
- Q.6** a) Explain compressive and flexural strength of concrete. **05**
b) Discuss durability and permeability of concrete. **05**
- Q.7** a) Explain transportation and compaction of concrete. **05**
b) Describe special concrete types and their applications. **05**